

The link between intellectual property rights, innovation, and growth: A meta-analysis

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ABSTRACT

In recent decades, intellectual property rights (IPR) and their macroeconomic effects have attracted considerable attention from both policymakers and academics. Despite a substantial theoretical and empirical literature, evidence regarding the impact of IPR protection on innovation and economic growth is mixed. In this paper we conduct a literature review and meta-analysis of the topic, and find that IPR have an overall positive effect on innovation and growth. However, the effect on innovation is weaker in developing countries than developed countries. We also find that the effects reported in the literature depend to a great extent on individual studies' methodological characteristics.

1. Introduction

Intellectual property rights (IPR) are widely believed to play a crucial role in encouraging innovation, fostering technological progress, and stimulating economic growth (Barro and Sala-i-Martin, 2004). As Machlup (1961) and Arrow (1962) argue, IPR encourage innovation, because they grant successful inventors temporary monopoly power over their innovations. Over the years, many IPR instruments with different purposes and application fields have been developed. From the Industrial Revolution onward, the power of preventing others from using one's intellectual creations has been strengthened to encourage private investment in innovation activities and promote sustainable economic growth. As Braga et al. (1999) emphasize, the main intention has been to regulate the relationship between innovators/creators and consumers and to reward innovators/creators for their ideas.

Nevertheless, many scholars have questioned whether IPR are in fact responsible for inducing innovation and, as a result, economic growth. How or how much IPR stimulate innovation and economic growth has

attracted considerable attention from economists. The related theoretical literature has increased considerably, and many empirical studies have been conducted to assess the effects of IPR on innovation and growth.¹

For example, Torun and Cicekci (2007); Arora et al. (2008); Ang et al. (2014); Chu et al. (2014); Yang et al. (2014); and Biel (2015) argue that IPR protection has a positive impact on innovation and growth. In contrast, Grossman and Helpman (1991) and Boldrin and Levine (2009), among others, find that IPR protection should be reduced in order to encourage creation and innovation, and thereby foster growth. Hence, as Cho et al. (2015) state, the causal relationship between IPR protection, innovation, and economic growth is somewhat ambiguous, in both theory and the empirical literature. The complexity of the effects across countries, firms, and time is justified by the variety of approaches in the literature. For example, Ginarte and Park (1997) conclude that IPR are important for R&D activities in developed economies, but not in developing economies. Chen and Puttitanun (2005) find a positive impact of IPR on innovation in developing countries and a U-shaped relationship between IPR and economic growth. Falvey et al. (2006a) show that

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¹ There is also a substantial body of literature that examines the impacts of IPR on other dimensions, such as the labour market (e.g., Mondal and Gupta, 2008; Zheng et al., 2020); FDI and trade (e.g., Yang and Cheng, 2008; Hsu and Tiao, 2015); inequality (e.g., Chu and Peng, 2011; Zhang et al., 2018); and productivity (e.g., Canavire-Bacarreza and Peñarrieta, 2020).

stronger IPR protection induces faster growth in developed countries and can also have a positive impact on the growth rate of the poorest countries. Thus, the effects of IPR are not completely clear.

Our motivation stems from the variation in results of empirical studies and the diversity of approaches used to assess the link between IPR, innovation, and growth. Our main research questions are: Why are the studies' conclusions so divergent? Given the empirical literature, is it possible to establish a consistent relationship between IPR, innovation, and growth?

We employ a meta-analysis to answer these questions. This technique uses statistical methods to summarize and analyse the empirical results, thus providing a more formal and objective process for reviewing empirical studies (Stanley, 2001). It also allows us to compare different research results and identify the sources of their heterogeneity.

We find that after correcting for publication bias, the overall effect of IPR on innovation and growth is positive. However, the effect on innovation is weaker in developing countries—where investment in innovative activities is low and it is preferable to imitate external innovations—than in developed countries, where the relevant policies and the economic and institutional environment are conducive to domestic innovation. We also find that the heterogeneity in the reported effects of IPR can be explained by differences in studies' methodological characteristics, such as the data structure, level of data/analysis, measurement of IPR, sample size, or publication quality. Our main findings have relevant policy implications.

The paper is organized as follows: Section 2 provides a brief review of the theoretical and empirical literature on the relationship between IPR, innovation, and economic growth. Section 3 presents the concepts and objectives of the meta-analysis, as well as the criteria used for their inclusion. Section 4 reports estimates of the average effect of IPR on innovation and growth and tests for the presence of publication bias. A meta-regression is estimated in Section 5 in order to identify the sources of variation in the reported effect sizes, and Section 6 concludes.

2. Literature review

2.1. Intellectual property rights: historical overview

In the contemporary world the main economic development challenge is based on the ability to generate and manage knowledge appropriability, and on the distribution of wealth. Over the years the concept of business wealth has changed and currently human and knowledge capital are widely regarded as the most valuable intangible assets. IPR in turn have become the base and the motivation for technological progress and new knowledge.

Indeed, the creation and dissemination of new knowledge, innovations, and social welfare have encouraged the implementation of a structured IPR regime that promotes creative and innovative activities through economic incentives and exclusive commercial rights for the creators (Adams, 2011). This kind of protection concept and rewarding emerged in the 4th century BC, with Aristotle's and Hippodamus of Miletus' debate. However, the present governmental IPR concept was born in the 13th century in Venice, with the need to protect the knowledge embedded in the glass industry and as a way to reward foreigners who brought new knowledge to the Venetian industry (e.g., Long, 1991). IPR protection then spread from Venice to other developed countries in Europe, the United States, Japan, South Korea, and Taiwan.

To harmonize and promote cooperation amongst states, IPR have undergone substantial changes and a vast number of international treaties and agreements have been signed for that purpose. The major agreement was the Trade-Related Aspects of Intellectual Property Rights (TRIPS), negotiated between 1986 and 1994 in the Uruguay round with the goal to induce more innovation that would accelerate economic growth, signed by 171 developed and developing countries and managed by the World Intellectual Property Organization (WIPO).

IPR regimes differ between developed and developing countries. In

developed countries the IPR regime protects all categories of intellectual properties with strong mechanisms, and has a well-structured administration to create conditions that compensate large R&D investments. The main goal is to promote economic growth by encouraging creative and innovative activities (Falvey et al., 2006b). In developing countries, only a few types of IPR are protected and those in only a rudimentary way. This kind of protection, or rather the lack of it, allows foreign developed firms to invest in developing countries promoting imitation. Imitation is the main source of technological development in these countries: the lower IPR protection seems to be the best way to encourage more domestic profitable activities, ensuring sustainable economic growth. The knowledge transfer from developed countries emerges as the base and the incentive source of domestic innovative activities.

In the same line, Chu et al. (2014) conclude that the effects of IPR on growth is stage dependent. In the framework of a Schumpeterian growth model of distance to frontier, the authors show that at an early development stage, it is optimal for countries to weaken IPR protection in order to promote imitation, while at later development stages countries tend to strengthen IPR protection in order to encourage domestic innovation. Thus, the optimal level of IPR protection increases as countries move towards the world technology frontier. This pattern is consistent with the evolution of IPR protection in China in recent decades.

IPR protection thus performs an important role among countries. However, it also presents some problems, since the reward criteria are based on market power and the welfare gain is often compromised (Gilbert and Shapiro, 1990). In order to make the spillovers of technological knowledge available to imitators, the scope and the duration of the protection should be restricted with the purpose of leading society to greater innovative economic development and prosperity – e.g., Kitch (1998), Towse and Holzhauser (2002), and Thumm (2000). The dissemination of new knowledge, further innovation, and growth could also be compromised by excessive IPR protection, since the spread of new ideas is limited by the excessive monopoly effects (Falvey et al., 2006a).

To sum up, as pointed out by Arrow (1962), in the absence of an IPR system, the monopoly value would be reduced or even extinguished, since the knowledge would become a public good,² and anyone could use it for a small (or any) cost (e.g., Maskus and Reichman, 2005). Thus, the trade-off is between allowing access to the knowledge and stimulating the investment in R&D activities.

Table 1 summarizes the overview performed on IPR.

2.2. IPR, innovation, and economic growth: an ambiguous relationship

Theoretical studies about IPR, innovation, and growth have been considered since Adam Smith, in 1776, although only with Robert Solow, in 1957, did they include the neoclassical growth models. Nevertheless, these models are not able to explain the factors behind long-run economic growth. Then, with Romer (1986) and Lucas (1988) a new growth theory emerged with endogenous technological change and human-capital formation. The new theory assumes that technological progress, which is the basis of societal transformation, globalization, and long-run growth, occurs through investment in innovative R&D activities carried out or supported by firms searching for profits (e.g., Barro and Sala-i-Martin, 2004).

As Arrow (1962) concluded, invention is a highly risky activity. Hence, the expected investment in R&D activities and growth are reduced without any kind of sure return. The IPR system appears then as a key factor for innovation and growth since the main purpose is to reward innovators for their creations. In line with Machlup (1961), a time-limited monopoly promotes growth in the register of property rights for knowledge goods, allowing the creator to exploit the invention

² That is, a good that is systematically underprovided by private market forces and for which such under-provision has important international externality effects (Mercurio, 2013).

Table 1
Intellectual Property Rights: an overview.

Authors	Argument about IPR
Adams (2011)	The creation and dissemination of new knowledge, innovation, and social welfare encouraged the creation of a structured IPR regime.
Mansfield et al. (1981)	The costs of imitation are lower than the costs of innovation, and IPR ensure crucial leverage power to promote the technological change.
Falvey et al. (2006a)	IPR can encourage creative and innovative activities; however, excessive IPR protection can also compromise the spread of new ideas.
Chu et al. (2014)	The strength of IPR protection should increase as a developing country goes forward toward the world technological frontier.
Gilbert and Shapiro (1990)	Since the IPR reward criteria are based on market power and the welfare gain, IPR are often compromised.
Bielig (2015)	The scope and the duration of IPR should be restricted to lead society to greater innovative economic development and prosperity.
Arrow (1962)	In the absence of an IPR system, the monopoly value would be reduced or even extinguished since knowledge would become a public good.

economically.

Torun and Cicekci (2007) consider that IPR have a master role in the resolution of the innovation and economic growth challenges. Similarly, Arora et al. (2008) find that IPR directly enhance R&D innovative activities, which in turn increase growth. Ang et al. (2014) and Yang et al. (2014) reach similar conclusions: IPR have a positive impact on innovation and on growth. IPR importance is also validated, for example, by the result of the multiple linear regression model conducted by Bielig (2015), who estimated the economic effect of four main IPR categories and concluded that IPR have had a positive impact on the German economy.

However, IPR can also be an obstacle to the diffusion of knowledge and reduce incentives for further innovation. The seminal work of Grossman and Helpman (1991), for example, stated that the observation/analysis of rival innovations could benefit the innovators' knowledge and expand it. In other words, strong IPR protection may not be the key to long-run leverage of innovation and growth. Similarly, Boldrin and Levine (2009) determined empirically that IPR protection should be reduced in order to encourage creation and innovation, and to thereby increase growth.

Studies that examine the effect of IPR protection on innovation and growth usually point to an overall positive effect, although most of the time and for the specific case of economic growth, the effect is not clear. It is thus understood that, as stated by Cho et al. (2015, p. 827), "the causal relationship among intellectual property rights (IPR) protection, innovation and economic growth remains ambiguous" in both theoretical and empirical literature. The economic literature suggests that the connection from IPR to innovation and growth is very complex and depends on a range of factors, sometimes contradictory. Thus, IPR protection may not have a direct effect on economic growth, affecting other variants that, in turn, affect economic growth (Kanwar and Evenson, 2003). The ambiguity led Cozzi (2009) to discuss a few studies, and based on Tetsugen Haryuyama's and Michele Boldrin's papers, concluded that innovation and growth can develop relatively well without IPR protection, and that it is possible to find evidence of both positive and negative effect of IPR on innovation and growth.

The ambiguity of the effects of IPR on innovation and growth across countries, firms, and time can be justified by the variety of approaches in the empirical studies. For example, in a context with developed and developing countries, Gould and Gruben (1996) examine the impact of IPR on growth, using cross-country data on patent protection, trade regime, and country-specific features. The first results suggested that IPR had a positive effect on economic growth. In order to solve the estimation measurement and endogeneity problems, Gould and Gruben (1996) used instrumental variables techniques. However, the solution drove the statistical significance to a level lower than it had been initially, raising doubts about the importance of IPR on economic growth. The results became positive and stronger only in more open economies, when the dependency of the trade regime of a country was examined.

Using a panel dataset of over 70 countries, Kim et al. (2012) found that patent protection is an important determinant of innovation and that patentable innovations contribute to economic growth in developed countries, but not in developing countries. Focusing on developing economies, Adams (2009) finds that strengthening IPR has a negative effect on growth. Ginarte and Park (1997) use an augmented Mankiw

et al. (1992) model and consider 60 countries from 1960 to 1990 to conclude that IPR protection remains nonsignificant in all growth equations. Moreover, "IPR matter for the R&D activities of the developed economies, but not for those of the less developed economies" (Ginarte and Park, 1997, p. 60), suggesting that in developing countries the most significant R&D activity is imitation. Lerner (2009) examines how changes in patent policy occurring in several countries over the last 150 years have affected innovation, and found a negative impact of patent protection-enhancing shifts.

Kanwar and Evenson (2003) focus their analysis on examining the technological change from 1981 to 1995 by using cross-country panel data on R&D investment, patent protection, and other country-specific features. Their results indicate that IPR stimulate innovation. Grossman and Lai (2004) show that in developed countries with a high level of globalization, strong innovation capability, large domestic market, and high-market concentration, stronger IPR protection is useful to increase innovation and economic growth. In developing countries, the IPR protection must be reduced to avoid the slowdown of the innovation process, economic growth, and the increase of firm costs due to lawsuits to defend domestic patents. In these countries with an underdeveloped R&D sector growth will not be enhanced by innovation since it is not important enough; it is enhanced by factors such as international trade or foreign direct investment (FDI). Using a panel data set including 64 developing countries over the 1975–2000 period, Chen and Puttitanun (2005) evaluate empirically the theoretical results of IPR and innovation in developing countries, confirming the positive impact of IPR on innovation in developing countries and a U-shaped relationship between IPR and economic growth.

Using panel data of 19 developed and 28 developing countries from 1970 to 1990 and to determine the country's rate of innovation and economic growth, Schneider (2005) examines the importance of high-technology trade, IPR, and FDI. Based on Grossman and Helpman's (1991) and Barro and Sala-i-Martin's (1997) models, Schneider (2005) estimates the innovation and growth rates using fixed effects (FE) and Ordinary Least Squares (OLS) regressions. Results suggest that the IPR level is an important factor to explain the innovation rate. Conversely, the growth rate is not explained by the IPR level, but by the stock of physical capital and foreign technology. Employing threshold regression techniques, Falvey et al. (2006a), estimate an equation of growth to examine the non-linear relationship between IPR and the level of development in a dataset of 79 countries from 1975 to 1994. Their results reveal that a stronger IPR protection scenario is expected to enhance growth in countries with high per capita income, and that this can also have a positive impact on the growth of the poorest countries. The relationship between IPR protection and growth in the poorest countries does not result from the encouragement of domestic R&D and innovation.

Hudson and Minea (2013) used a Panel Smooth Threshold Regressions model created by Gonzalez et al. (2005) to perform the estimation of IPR level on innovation, using a dataset of 62 developed and developing countries. They concluded that IPR exert a complex effect on innovation and that the relationship is essentially a U-shaped curve. In a panel of 72 developed and developing countries and 54 manufacturing industries from 1981 to 2000, Hu and Png (2013) investigate the impact of changes in effective patent rights and concluded that IPR protection

Table 2
The linkage between IPR, Innovation, and Growth.

Author(s)	Argument	Relationship between IPR and	
		Innovation	Growth
Arrow (1962)	IPR display an important role on R&D activities	Positive	Positive
Torun and Cicekci (2007)	IPR have a master role in the resolution of innovation and economic growth challenges.	Positive	Positive
Arora et al. (2008)	IPR reward innovators for their creations and directly enhance innovation through the investment in R&D activities, which in turn increases growth.	Positive	Positive
Yang et al. (2014)	The role of IPR protection in stimulating innovations across countries.	Positive	Positive
Bielig (2015)	IPR had a positive impact on the German economy, reflecting the strong economy structure and innovation system.	Positive	Positive
Grossman and Helpman (1991)	IPR protection may not be the key to long-run leverage of innovation and economic growth.	Negative	Negative
Boldrin and Levine (2009)	IPR protection should be reduced to increase growth and encourage creation and innovation.	Negative	Negative
Gould and Gruben (1996)	Results point to doubts about the importance of IPR on economic growth.	N/A	Inconclusive
Adams (2009)	IPR have a negative impact on economic growth in developing countries.	N/A	Negative
Kim et al. (2012)	Patent protection is a determinant of innovation and patentable innovations contribute to economic growth in developed countries, but not in developing countries.	Positive	Undefined
Ginarte and Park (1997)	IPR protection is insignificant for innovative activities and for economic growth.	Insignificant	Insignificant
Grossman and Lai (2004)	IPR protection is useful to increase innovation and economic growth, but only in developed countries.	Undefined	Undefined
Lerner (2009)	Patent protection-enhancing policies have a negative impact on innovation.	Negative	N/A
Kanwar and Evenson (2003)	IPR protection encourages innovation and may not have a direct effect on economic growth.	Positive	Indirect
Schneider (2005)	IPRs affect the innovation rate, but this impact is more significant for developed countries.	Positive	Indirect
Chen and Puttitanun (2005)	There is a positive impact of IPRs on innovations in developing countries and a U-shaped relationship between IPRs and economic development.	Positive	U-shaped
Falvey et al. (2006a)	The effect of IPR on growth is positive and significant for low- and high-income countries, but not for middle-income countries.	Positive	Positive
Hu and Png (2013)	IPR protection has a positive impact on industrial and economic growth even though it is stronger in developed countries than in developing ones.	N/A	Positive
Hudson and Minea (2013)	IPR exert a complex effect on innovation and the relationship is essentially a U-shaped curve.	U-shaped	N/A
Mrad (2017)	IPR protection positively affects economic growth in developing countries by attracting foreign technology embodied in capital goods.	N/A	Positive
Thompson and Rushing (1999)	Patent protection has a positive effect on total factor productivity and economic growth in richer countries.	N/A	Positive
Sweet and Maggio (2015)	Stronger IPR engender higher levels of economic complexity, but only in countries with an initial above-average level of development and complexity.	Positive	N/A
Zhang et al. (2015)	The weak IPR–growth evidence in previous studies may be due to a neglect of the role of finance markets and private IPR.	Positive	Indirect
Varsakelis (2001)	Countries with a strong patent protection framework tend to invest more in R&D	Positive	N/A
Sakakibara and Branstetter (2001)	Japanese firms did not increase significantly their R&D effort and innovative output in response to the country's 1988 patent reform.	Insignificant	N/A
Branstetter et al. (2011)	US-based MNE that make extensive use of IPR increased their use of inputs and expanded their scale of activities in response to IPR reforms.	N/A	Positive
Ang et al. (2014)	Local enforcement of IPR has an important role in encouraging financing and investing in R&D.	Positive	Positive
Bielig (2015)	Different types of IPR had different effects on German GDP.	N/A	Ambiguous

has a positive impact on industrial and economic growth even though it is stronger in developed countries than in developing ones. Sweet and Maggio (2015), Zhang et al. (2015), Mrad (2017), and Thompson and Rushing (1999) also found positive and strong effects of IPRs on innovation and growth.

Varsakelis (2001) examines the effect on R&D investment of national culture, openness of the economy, and patent protection framework, finding that the national culture and the degree of openness are correlated with R&D investment and that countries with a strong patent protection framework tend to invest more in R&D.

While most of the studies use country-level data, some examine the IPR-innovation-growth link in a specific country using firm- or industry-level data. For example, Sakakibara and Branstetter (2001) study firm responses to the Japanese patent reforms of 1988 that expanded the scope of patent rights, and find that the increases in R&D effort and innovative output were quite modest. Branstetter et al. (2011) analyse the responses of US-based multinationals and domestic industrial production to a set of IPR reforms in the 1980s and 1990s. They find that multinationals that make extensive use of IPR disproportionately increased their use of inputs and expanded their scale of activities. Using a database of high-technology firms in China, Ang et al. (2014) show that effective enforcement of IPR at the provincial level plays an important role in encouraging financing and investment in R&D. Bielig (2015) analyses the relationship between the registered IPR and economic development in Germany and shows that different types of IPR had

different effects on German GDP in the period 1999–2009 – while the stock of patents, trademarks, and designs had a positive impact, applications for utility models had a negative impact.

Table 2 summarizes the main results in this subsection.

To sum up, the literature analysed suggests that the effects of IPR are not completely clear. Most often the conclusions present a direct and positive impact of IPR on innovation and growth. However, the magnitude and the direction of the effect seem to depend on countries' structural characteristics, such as the development level and the openness degree of the economy. In addition, there are significant methodological differences amongst studies that may further contribute to explaining the heterogeneity in the empirical findings.

First, there are differences in data levels (country-level, industry-level or firm-level data), which may lead to different estimated effects of IPR on innovation and growth. Also, as emphasized by Moser (2013) and Hall et al. (2014), most analyses do not account for a direct link between products and the different protection mechanisms, which is an important shortcoming of the literature.

Second, studies use different indicators to measure the degree of IPR protection. Several indexes of IPR have been created (e.g., Rapp and Rozek, 1990; Sherwood, 1997; Ostergard, 2000; Liu and LaCroix, 2015), the most common being the Ginarte-Park Index (GPI). The GPI was initially developed by Ginarte and Park (1997) and later updated and extended by Park (2008) and Park and Lippoldt (2008). It is an unweighted sum of the values attributed to five components of IPR in each

country: the extent of coverage of patent protection; membership in international patent agreements; provisions for loss protection; enforcement mechanisms; and duration of protection. Despite being the most commonly used measure of IPR in the empirical literature, the GPI does not capture all the dimensions of IPR and thus should be complemented with other measures when examining the IPR-innovation-growth relationship. For example, there are innovations that are patentable but not patented (usually because firms perceive higher returns from keeping ideas as trade secrets), and there are innovations that are not patentable but could be secured with other formal property rights (such as copyright) – as in e.g., [Chen and Puttitanum \(2005\)](#) and [Moser \(2005, 2012, 2013\)](#).³ Moreover, the GPI does not capture the enforcement of IPR laws, which according to some authors is relevant for explaining the IPR-innovation link.⁴

Finally, it is also possible that patents and other formal protection instruments such as copyright do not in fact encourage investment in R&D as much as might be expected. For example, [Jones and Williams \(1998\)](#) and [Bloom et al. \(2013\)](#) note that in the USA the optimal investment in R&D is higher than the actual investment. Thus, despite the good conditions in the USA for the definition and enforcement of intellectual property rights, there is underinvestment in R&D and, therefore, the rates of innovation and economic growth are below the country's potential. Other works (e.g., [Hu and Png, 2013](#)) have noted that patents are not fulfilling their objectives of fostering innovation and economic growth. In this context, a recently emerged strand of literature is concerned with alternative mechanisms to encourage innovation, so as to include government purchase of patent rights ([Kremer, 1998](#)), awards ([Shavell and Ypersele, 2001](#)), and rewards as a percentage of sales revenues ([Grinols and Lin, 2011](#)). More generally, however, academics seem to be increasingly interested, on the one hand, in the role of open source agreements (on property rights) – as in e.g., [Boyle \(2003\)](#) and [Lerner and Tirole \(2005\)](#) – and, on the other hand, in the role of what [Hall et al. \(2014\)](#) and [Sweet and Maggio \(2015\)](#) refer to as informal intellectual property.⁵

Given the complexity of the relationship between R&D effort, resulting innovations, economic growth, and IPRs, in the next sections we develop a meta-analysis on the literature that estimates the impact of IPR on innovation and on economic growth. The main goal is to detect some potential factors that determine the differences in studies' findings and thus have a better understanding of the nature of the IPR-innovation-growth links.

3. Concept and objectives of meta-analyses and studies' selection

3.1. Meta-analysis: concepts and objectives

As stated by [Glass \(1976, p.3\)](#), meta-analysis is “the analysis of analyses”. It is a methodology that “provides a more formal and objective process on reviewing empirical literature. It employs conventional statistical methods and criteria to summarize and evaluate empirical economics” ([Stanley, 2001, p.147–148](#)). It is a method that synthesizes, combines, and explains the results of the empirical literature on a specific topic ([Nijkamp and Poot, 2004](#)).

³ We thank an anonymous referee for the comments and suggestions on this topic.

⁴ For example, [Park \(2005\)](#) finds that what matters most about IPR and their effects on R&D activities are the mechanisms for enforcement and the level of enforcement effectiveness, especially for developing countries. [Papageorgiadis and Sharma \(2016\)](#) also find that enforcement is significant in explaining the relationship between innovation and the IPR system. In the same line, [Ang et al. \(2014\)](#) capture not just patents rights and their infringement but also commercial secrets and infringements, and other kinds of IPR infringements, which the literature recognizes to be important determinants of innovative activity.

⁵ We thank an anonymous referee for the comments and suggestions on this topic.

Although it was first used in medicine and psychology, meta-analysis rapidly expanded to other research fields. In economics it has been widely employed, especially to examine those topics in which results are not consensual (e.g., [Card and Krueger, 1995](#); [Gorg and Strobl, 2001](#); [Kremers et al., 2002](#); [Doucouliagos and Paldam, 2008](#); [Iamsiraroj and Ulubasoglu, 2015](#); [Anderson et al., 2018](#)).

Meta-Regression Analysis is a specific technique used in meta-analysis that seeks to explain the heterogeneity in the effect sizes reported in the primary studies. It involves the estimation of a standard regression model as follows (e.g., [Stanley and Jarrell, 1989](#)):

$$b_j = \beta + \sum_{k=1}^K \alpha_k Z_{kj} + e_j, (j = 1, 2, \dots, L) \quad (1)$$

where: b_j is the reported estimate of the phenomenon β of interest in study j in a literature comprised of L studies; Z_{kj} corresponds to the K meta-independent variables that measure relevant characteristics of the empirical studies (e.g., estimation techniques, sample period, data structure); α_k are the meta-regression coefficients that complement the studies' characteristics on b_j ; and e_j is the regression error. Equation (1) can be estimated to evaluate how the studies' characteristics influence the estimation of the effect sizes.

3.2. Studies' selection for the meta-analysis and definition of the effect size

Before the application of the meta-analysis it is important to perform a careful selection of the papers that are included in it. In the first stage we used the Web of Science and searched in the title and in the abstract of published papers and working papers for any reference to “Intellectual Property Rights” and “Innovation” and “Growth”, “IPR and Innovation and Growth”, “IPR and Innovation”, “IPR and Growth”, “IPR and R&D and Growth”. The search led to more than 1000 papers in total, of which approximately 400 were selected from the field of “economics”. We reviewed the empirical studies that present estimates of the impact of IPR on innovation and growth. From these, we had to exclude those that do not report the necessary information to support the meta-analysis. We were left with a final set of 22 studies (listed in [Table 3](#)).

The summary measures that we use in the meta-analysis are the coefficients reported in the primary studies for estimating the effect of IPR on innovation and growth. Given that the primary studies use different measures of IPR, innovation and growth, it is necessary to use a metric that makes data comparable. Following [Ugur \(2014\)](#), we use the partial correlation coefficient (PCC), which is our effect size. The PCC, r_i , and its standard errors, se_{ri} , are calculated based on the following equations:

$$r_i = t_i / \sqrt{t_i^2 + df_i} \quad (2)$$

and

$$se_{ri} = \sqrt{(1 - r_i^2)/df_i} \quad (3)$$

where t_i represents the t -statistic associated with each coefficient estimate reported in the primary study, and df_i represents the degrees of freedom used in the respective estimation. We collected from each paper all of the reported coefficients and respective standard errors. This resulted in a total of 256 estimates of the IPR impacts on Innovation and 206 estimates of the IPR impacts on Growth. [Table 4](#) presents some characteristics of each paper.

The estimates of the IPR effects on Innovation vary between a minimum of -0.7129 ([Sweet and Maggio, 2015](#)) and a maximum of 0.8240 ([Yang et al., 2014](#)), with 44 negative values and 212 positives. 114 of the 256 estimates were obtained using a mixed sample of developing and developed countries, meaning that the remaining 142 used either one or the other group of countries. The majority of the estimates (199) were obtained using panel data.

Table 3
Studies selected for the meta-analysis.

Authors	Title	Journal
Adams (2009)	Intellectual property rights, political risk and economic growth in developing countries	Journal of Economics and International Finance
Ang et al. (2014)	Does enforcement of intellectual property rights matter in China? Evidence from financing and investment choices in the high-tech industry	Review of Economics and Statistics
Branstetter et al. (2011)	Does intellectual property rights reform spur industrial development?	Journal of International Economics
Chen and Puttitanun (2005)	Intellectual property rights and innovation in developing countries	Journal of Development Economics
Chu et al. (2014)	Stage-dependent intellectual property rights	Journal of Development Economics
Falvey et al. (2006a)	Intellectual property rights and economic growth	Review of Development Economics
Gould and Gruben (1996)	The role of intellectual property rights in economic growth	Journal of Development Economics
Hu and Png (2013)	Patent rights and economic growth: evidence from cross-country panels of manufacturing industries	Oxford Economic Papers
Hudson and Minea (2013)	Innovation, Intellectual Property Rights, and Economic Development: A Unified Empirical Investigation	World Development
Kim et al. (2012)	Appropriate intellectual property protection and economic growth in countries at different levels of development	Research Policy
Lerner (2009)	The Empirical Impact of Intellectual Property Rights on Innovation: Puzzles and Clues	American Economic Review
Mrad (2017)	The Effects of Intellectual Property Rights Protection in the Technology Transfer Context on Economic Growth	Journal of Innovation Economics & Management
Kanwar and Evenson (2003)	Does intellectual property protection spur technological change?	Oxford Economic Papers
Park and Ginarte (1997)	Intellectual property rights and economic growth	Contemporary Economic Policy
Park (2005)	Do Intellectual Property Rights Stimulate R&D and Productivity Growth? Evidence from Cross-national and Manufacturing Industries Data	<i>in</i> Intellectual Property Rights and Innovation in the Knowledge-Based Economy, J. Putnam (Ed.)
Sakakibara and Branstetter (2001)	Do Stronger Patents Induce More Innovation?	The RAND Journal of Economics
Schneider (2005)	International trade, economic growth and intellectual property rights: A panel data study of developed and developing countries	Journal of Development Economics
Sweet and Maggio (2015)	Do Stronger Intellectual Property Rights Increase Innovation? Evidence from the 1988 Japanese Patent Law Reforms	World Development
Thompson and Rushing (1999)	An Empirical Analysis of the Impact of Patent Protection on Economic Growth: An Extension	Journal of Economic Development
Varsakelis (2001)	The impact of patent protection, economy openness and national culture on R&D investment: a cross-country empirical investigation	Research Policy
Yang et al. (2014)	Do stronger intellectual property rights induce more innovations? A cross-country analysis	Hitotsubashi Journal of Economics
Zhang et al. (2015)	How Private Property Protection Influences the Impact of Intellectual Property Rights on Economic Growth?	Global Economic Review

As for the estimates of the IPR effects on Growth, the minimum value is -0.3142 (Park, 2005) and the maximum is 0.7402 (Zhang et al., 2015). Of the 206 estimates collected, 169 assume positive values, and 128 were obtained using a mixed sample of developing and developed countries. The sample contains 106 estimates from panel studies and 100 from cross-section studies.

4. Publication bias and estimation of average effect

One of the first steps in conducting a meta-analysis is to test whether there is publication bias. In its most common form publication bias refers to the tendency for authors to produce and editors to publish only statistically significant results. This leads to a distortion in the results reported by the empirical literature on a specific topic (Doucouliagos, 2005).

In a simple way, this issue can be examined using a funnel plot; that is, a scatter figure that shows the relationship between the effect sizes on the horizontal axis (the partial correlation coefficient between IPR and innovation/growth) and the inverse value of the estimate of its standard error, as a measure of precision, on the vertical axis (e.g., Neves et al., 2016). In such figure the absence of publication bias is characterized by a symmetrical inverted funnel shape, with values varying randomly and symmetrically around the average “true effect” (Benos and Zotou, 2014). But if publication bias exists, the *funnel plot* should be asymmetrical, especially at the bottom. Thus, the (a)symmetry of the funnel allows us to assess the existence of publication bias (Stanley, 2005). Figs. 1 and 2 are the funnel plots for the reported estimates of IPR effects on innovation and growth, respectively.

Visual inspection of Figs. 1 and 2 suggests that the average effect of IPR on innovation and growth is positive. Moreover, there are some traces of publication bias, especially in the IPR-growth estimates, as the second funnel plot seems to be asymmetrical toward the right side.

The funnel plot can be formally tested using the Funnel Asymmetry Test (FAT), which consists of running a simple regression of the effect sizes on the respective standard errors, as in the equation below:

$$r_i = \alpha_0 + \alpha_1 SE_{ri} + \mu_i \quad (4)$$

In the presence of publication bias, authors of studies with higher standard errors are compelled to look more intensively for higher estimates of the effect size, in an attempt to report statistically significant results. Thus, if publication bias exists, r_i will be correlated with standard errors SE_{ri} ; if no bias exists, the reported estimates will vary randomly around the effect α_0 , regardless of the value of SE_{ri} (Stanley, 2005).

However, equation (4) presents a heteroscedasticity problem, which can be solved by dividing both sides by SE_{ri} (Stanley, 2005), yielding:

$$t_i = \alpha_0 precision_{ri} + \alpha_1 + e_i \quad (5)$$

where $precision_{ri} = 1/SE_{ri}$ and $t_i = \varphi_i/SE_i$ is the t -statistic associated with φ_i reported in the primary studies. A test for $\alpha_1 \neq 0$ in equation (5) is thus a test for the presence of publication bias (FAT), while the estimate of α_0 is the estimate of the average effect size beyond publication bias (PET) – Ugur et al. (2016), Sequeira and Neves (2020).

A second econometric problem in estimating equation (5) by OLS is the presence of statistical dependence. Statistical dependence exists because the observations taken from the same study, sharing the same

Table 4
Main characteristics of the primary papers.

Article	Impact on	NE	PCC IN	SD PCCIN	PCC GR	SD PCCGR	Type of countries	Data structure	Data level	Estimation method	Measure of IPR
Adams (2009)	Growth	3	−0.1596	0.0755			Developing	Panel	Country	SUR	GPI
Ang et al. (2014)	Innovation	20	0.0054	0.0086			Developing	Panel	Firm	Poisson	IPR index created by the authors
Branstetter et al. (2011)	Growth	10			0.0370	0.0102	Developing	Panel	Firm	OLS	IPR reforms
Chen and Puttitanun (2005)	Innovation	4	0.1324	0.0798			Developed	Panel	Industry	OLS	GPI
Chu et al. (2014)	Growth	3			−0.1001	0.0435	Developing	Panel	Country	FE	GPI
Falvey et al. (2006a)	Growth	18			0.0941	0.0648	Developed	Panel	Country	2SLS	GPI
Gould and Gruben (1996)	Growth	4			0.1787	0.1057	Developed	Cross Section	Country	Threshold	GPI
Hu and Png (2013)	Growth	16			0.0721	0.0392	Developed	Panel	Industry	OLS	RRI
Hudson and Minea (2013)	Innovation	7	0.4178	0.0470			Developed	Panel	Country	IV	MIPRI
Kanwar and Evenson (2003)	Innovation	9	0.3169	0.1220			Developed	Panel	Country	PSTR	GPI
Kim et al. (2012)	InnovationGrowth	24	−0.0257	0.0643	0.0605	0.0395	Developing	Panel	Country	RE	GPI
Lerner (2009)	Innovation	1	0.0277	0.0793			Developed	Panel	Firm	OLS	Utility Models
Mrad (2017)	Growth	6			0.0264	0.0531	Developing	Panel	Country	FE	GPI
Park and Ginarte (1997)	InnovationGrowth	8	0.3954	0.1388	−0.0700	0.1554	Developing	Cross-Section	Country	SGMM	Length of patent protection
Park (2005)	InnovationGrowth	64	0.1919	0.1190	0.0607	0.1101	Developed	Cross-Section	Country	2SLS	GPI
Sakakibara and Branstetter (2001)	Innovation	10	0.0484	0.0199			Developing	Cross-Section	Industry	OLS	8 different measures of IPR
Schneider (2005)	InnovationGrowth	54	0.1230	0.1040	0.1010	0.1029	Developed	Panel	Firm	RE	1988 Patent reform in Japan
Sweet and Maggio (2015)	Innovation	43	0.0609	0.0469			Developed	Panel	Country	FE	GPI
Thompson and Rushing (1999)	Growth	3			0.1228	0.1671	Developed	Panel	Country	OLS SGMM	GPI
Varsakelis (2001)	Innovation	2	0.5118	0.1210			Developing	Cross Section	Country	SUR	RRI
Yang et al. (2014)	Innovation	45	0.2368	0.1149			Developed	Panel	Country	OLS	GPI
Zhang et al. (2015)	InnovationGrowth	76	0.3818	0.1268	0.1466	0.1382	Developing	Cross Section	Country	FE	IMDI
							Developed			RE Threshold	WEFI
										OLS	GPI
											MIPRI

Notes: **NE** means the number of estimates; **PCCIN** means the average of the partial correlation coefficients of IPR on Innovation; **SEPPCCIN** means the average of the standard errors of the partial correlation coefficients of IPR on Innovation; **PCCGR** means the average of the partial correlation coefficients of IPR on Growth; **SDPCCGR** means the average of the standard errors of the partial correlation coefficients of IPR on Growth; **OLS** means Ordinary Least Squares; **2SLS** means Two-Stage Least Squares; **RE** means random effect; **FE** means fixed effect; **IV** means instrumental variables; **PSTR** means Panel Smooth Threshold Regression; **SGMM** means System-GMM; **SUR** means Seemingly Unrelated Regression; **GPI** means Ginarte-Park Index; **RRI** means Rapp-Rozek Index; **WEFI** means World Economic Forum Index; **IMDI** means Institute for Management Development Index; **MIPRI** means Modified Intellectual Property Rights Index.

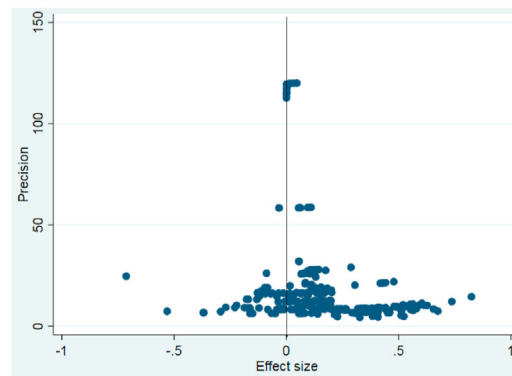
datasets, specifications, or estimation procedures, are likely to be correlated (Nelson and Kennedy, 2009). In this case, traditional OLS estimates are not appropriate, as they tend to produce biased estimates and exacerbate the sample size due to considering all estimates as independent observations (Snijders and Bosker, 2012; Ugur et al., 2020). A simple way to address this issue is to choose only one estimate from each study (Stanley, 2001). However, this is not desirable when the number of studies included in the meta-analysis is relatively small. If several estimates from each study are to be used in the meta-analysis, clustered data analysis, bootstrapped standard errors, panel data estimators, and hierarchical models can be used to address the problem of statistical dependence (Nelson and Kennedy, 2009; Doucouliagos and Laroche, 2009; Neves and Sequeira, 2018).

We opt to estimate equation (5) using hierarchical models (HLM). As

in the case of clustered standard errors techniques, HLM allow correcting the standard errors for within study-study correlation, as it considers that observations are clustered by study, each study representing a cluster. In addition, HLM allow estimating the model's coefficients considering the presence of heterogeneity between studies. When unobserved heterogeneity exists, HLM are preferred to OLS (Katahira, 2016).

In hierarchical models, observations are grouped according to their characteristics, and both within and between groups variation can be the cause for differences in individual observations. Since the model's coefficients are allowed to vary randomly between groups, a typical HLM can be written as:

$$Y_{ij} = (\beta_0 + \gamma_{0j}) + (\beta_1 + \gamma_{1j})X_{ij} + \varepsilon_{ij}, \quad (6)$$



Note: The horizontal axis represents the partial correlation coefficient between IPR and Innovation, and the vertical axis represents the inverse value of the estimate of its standard error as a measure of precision.

Fig. 1. IPR effect on Innovation – Funnel Plot. Note: The horizontal axis represents the partial correlation coefficient between IPR and Innovation, and the vertical axis represents the inverse value of the estimate of its standard error as a measure of precision.

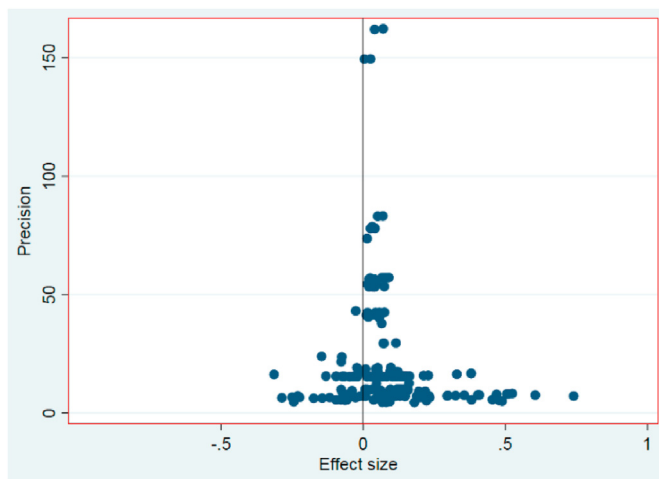


Fig. 2. IPR effect on Growth – Funnel Plot. Note: The horizontal axis represents the partial correlation coefficient between IPR and Growth, and the vertical axis represents the inverse value of the estimate of its standard error as a measure of precision.

where: subscript i represents the observation and j represents the group; β_0 is the fixed-effects intercept; β_1 the fixed-effects slope; γ_{0j} is the group-specific random intercept; and γ_{1j} is the group-specific random slope. Both γ_0 and γ_{1j} have a normal distribution with a mean equal to zero and a constant variance. This generic version of HLM is called the random-intercepts-and-slopes model. If there is heterogeneity between groups in the intercepts only, then $\gamma_{1j}=0$ and we have a random-intercepts-only model. If heterogeneity is present only in slopes, then $\gamma_{0j}=0$ we have a random-slopes-only model (Ugur et al., 2020).

The structure of HLM models can be used meta-analyses, since the observations (estimates of the effect size) are grouped by studies. In the context of meta-analysis, within-group variation corresponds to differences in the estimates of the effect size taken from the same study and between-group variation corresponds to differences across studies. Some examples of recent meta-analyses in economics literature using hierarchical models are Ugur et al. (2016), Neves and Sequeira (2018), Afonso et al. (2020), Ugur et al. (2020).

Tables 5 and 6 report the results of the estimation of equation (5) with the hierarchical specification presented in (6). Estimations were performed by restricted maximum likelihood.

The upper part of Tables 5 and 6 presents the estimates of the coefficients of the FAT-PET regression. The middle part reports the variances of the random components of coefficients and of the residuals. The

estimations confirm the results suggested by visual inspection of the funnel plots. First, the coefficient associated with precision is positive and statistically significant in both estimations, which means that, on average, IPR have a positive effect on innovation and growth. Second, the constant is also positive and statistically different from zero in the second regression, confirming the existence of publication bias in the IPR-growth literature, i.e., the tendency for journals to give preference to studies with positive and statistically significant results of IPR effect on growth.

Another important question in empirical economics research is whether the evidence base has an adequate statistical power. Adequately powered estimates are those that are able to detect a true empirical effect, in the event that one exists. Ioannidis et al. (2017) propose an estimator of the effect sizes based on an unrestricted Weighted-Least Squares (WLS) average that uses only the adequately powered estimates. An estimate is considered adequately powered when the respective standard error is smaller than the absolute value of the true effect divided by 2.8 (for more details, see Ioannidis et al., 2017). Thus, this weighted average of adequately powered (WAAP) estimator uses the most informative and less biased estimates, rather than all the estimates collected in the meta-analysis.

We have computed the WAAP estimators for our meta-analysis, considering as the true effect the coefficient associated with *Precision* reported in Tables 5 and 6, and the inverse of the standard errors as the weight of each adequately powered estimate. Of the 256 primary estimates of the IPR-innovation effect, 159 (62%) are adequately powered. The WAAP estimate of the impact of IPR on innovation is positive (0.0224) and statistically different from zero at 1%, but smaller than the average effect size reported in Table 5. As for the growth effect of IPR, only four estimates are adequately powered, which reveals that such effect is estimated with low power in the primary empirical studies. Surprisingly, this is not an uncommon situation, since around 20% of the areas of empirical economics do not contain a single estimate with an adequate power (Ioannidis et al., 2017).

5. Multivariate Meta-Regression Analysis

In this Section we estimate a multivariate meta-regression to assess how the differences in studies' methodological characteristics explain the heterogeneity in effect sizes. We pay particular attention to the differences highlighted in the Literature Review section. We account for differences regarding the: development level of the countries included in the sample of the primary studies; way of measurement of IPR; level (unit) of analysis; structure of the data; estimation techniques; sample size; moderator variables in the primary regressions; impact of the study in the academic community; and year of publication. The variables that capture these dimensions are described in Table 7.

Table 5Estimation of equation (5), for innovation – dependent variable, t .

Coefficients for:	
<i>Precision</i>	0.3074*** (0.1012)
<i>Constant</i>	–1.1783 (1.0701)
RE variances	
<i>Var (Precision)</i>	0.0785 [0.0181; 0.3405]
<i>Var (Constant)</i>	4.9358 [0.8056; 30.2402]
<i>Var(residuals)</i>	4.7701 [3.9164; 5.8099]
<i>N. obs (N. Studies)</i>	256 (15)
<i>Log restricted-likelihood</i>	–595.1160
<i>Wald test ($\alpha_0=0$)</i>	9.22***

Notes: Standard errors for coefficient estimators are in parentheses (). 95% confidence intervals for random effects variances are in brackets []. Level of significance: *** for p -value<0.01; ** for p -value<0.05; * for p -value<0.1. N. Obs. is the number of observations; N. studies is the number of studies. RE variances mean Random Effects variances.

Table 6Estimation of equation (5), for growth – dependent variable: t .

Coefficients for:	
<i>Precision</i>	0.0336*** (0.0052)
<i>Constant</i>	0.4594** (0.2184)
RE variances	
<i>Var (Precision)</i>	–
<i>Var (Constant)</i>	0.1003 [0.0004; 22.9681]
<i>Var(residuals)</i>	2.6579 [2.1541; 3.2795]
<i>N. obs (N. Studies)</i>	206 (12)
<i>Log restricted-likelihood</i>	–399.7772
<i>Wald test ($\alpha_0=0$)</i>	41.16***

Notes: See Table 5.

A frequent problem in multivariate meta-regressions with many potential covariates is model uncertainty, which is uncertainty regarding the choice of the covariates to be included. Some studies use Bayesian Model Averaging (BMA) to address model uncertainty. In the BMA

Table 7

Moderating variables.

Variable	Type	Measure
Nobs	Count	Number of observations (divided by 1000).
Year	Count	Year of publication – 1996.
Q1Journal	Dummy	1 if the study was published in a journal of the first quartile of the impact factor ranking; 0 otherwise.
Ncitations	Count	Number of citations received by the study in Google Scholar.
Ncountries	Count	Number of countries included in the sample.
Developing	Dummy	1 if the sample includes only developing countries; 0 otherwise.
Developed	Dummy	1 if the sample includes only developed countries; 0 otherwise.
Panel	Dummy	1 if panel data are used; 0 otherwise.
IV	Dummy	1 if instrumental variables are used to account for endogeneity; 0 otherwise.
Trade	Dummy	1 if trade openness is included as moderator variable; 0 otherwise.
GPI	Dummy	1 if IPR are measured by the Ginarte-Park Index; 0 otherwise.
Country	Dummy	1 if the paper uses country-level data; 0 otherwise.
Industry	Dummy	1 if the paper uses industry-level data; 0 otherwise.

approach several regressions are estimated with different combinations of the potential covariates. A weighted average of all regressions is then computed, with weights determined by goodness of fit and model complexity. The decision regarding which covariates to include in the meta-regression relies on the value of the Posterior Inclusion Probabilities (PIP), which measure the relevance of each covariate. BMA has been used in several meta-analyses in economics (e.g., Iršová and Havránek, 2013; Cuaresma et al., 2014; Churchill and Yew, 2017; Cardoso et al., 2020; Havránek and Sokolova, 2020).

In a recent meta-analysis, Ugur et al. (2020) address model uncertainty employing a WALs technique that combines elements from BMA and frequentist approaches. The authors argue that the WALs is preferred to the BMA because it is less time consuming in terms of computation, presents a bounded risk profile, and is more transparent in the focus variable selection. We use the WALs approach to determine the covariates to be included in our meta-regressions. Following Ugur et al. (2020), we: i) consider *Precision* as the focus variable; ii) consider all the variables presented in Table 7 as auxiliary covariates in the WALs estimation; iii) include in the main estimation of the meta-regression only those covariates for which the t -statistic is equal to or greater than 1 (in absolute value) in the WALs estimation. The application of this criterion led to the exclusion of variables *Year*, *Citations*, *Developed*, and *Trade* in the regression for Innovation and *Developing*, *Developed*, *IV*, and *Trade* in the regression for Growth.

The estimation results of the meta-regression including the selected covariates are shown in Tables 8 and 9.⁶ In column (1) the baseline regression is presented, in which the hierarchical specification is estimated by restricted maximum-likelihood. Columns (2), (3), and (4) present robustness estimations: column (2) estimates the same model using maximum-likelihood; column (3) excludes the outliers;⁷ and column (4) estimates the model by OLS with clustered-robust standard errors. In all cases the moderator variables are divided by the standard errors of the effect sizes, *SEr*.

The results reported in Table 8 show that the effect of IPR on innovation depends crucially on a country's level of development. Dummy *Developing* has a negative and highly significant effect in all four regressions, meaning that innovation impact of IPR is weaker in developing countries. This is an expected result, since IPR protection policies in developing countries are weak or non-existent and investment in innovative activities is low, and it is preferable to imitate external innovations (e.g., Falvey et al., 2006a; Chu et al., 2014). On the contrary, the policies adopted and the institutional and economic environment in rich countries tend to be conducive to domestic innovation and, consequently, the protection of innovations (e.g., Adams, 2011). From Table 8 we can also conclude that differences in the structure of the data, measure of IPR, level of data/analysis, and sample size are also relevant in explaining the heterogeneity of the effect sizes, as variables *Panel*, *GPI*, *Industry*, and *Nobs* are statistically significant in at least two regressions.

As for Table 9, the most significant variables are *Industry*, *Panel*, *Nobs*, *Q1Journal*, *Ncitations*, and *Year*. This means that as far as the growth effects of IPR are concerned there are important differences in the studies' results according to their data level and structure, sample size, quality, and time of publication.

6. Conclusions

In this paper we examined the empirical trends in the relationship between IPR, innovation, and growth. We began by emphasizing the

⁶ The results of the WALs routine are reported in the Appendix (Tables 10 and 11). We also report (in Tables 12 and 13) the results we would obtain if we used as covariates all of the variables presented in Table 7.

⁷ An observation is considered an outlier if it is below the lower limit $Q1 - 1.5(Q3 - Q1)$ or above the upper limit $Q3 + 1.5(Q3 - Q1)$, where $Q1$ and $Q3$ are the first and third quartiles, respectively.

Table 8

Estimation of multivariate meta-regression for innovation – dependent variable: t.

	Baseline Meta-regression	Robustness estimations		
		Maximum Likelihood	No outliers	OLS with clustered robust SE
	(1)	(2)	(3)	(4)
Coefficients for:				
<i>Nobs</i>	0.0098 (0.0062)	0.0105* (0.0058)	0.0084* (0.0050)	0.0236** (0.0112)
<i>Q1Journal</i>	−0.0833 (0.1095)	−0.0838 (0.0892)	−0.0566 (0.0730)	−0.1136 (0.1216)
<i>Ncountries</i>	0.0005 (0.0011)	0.0003 (0.0010)	−0.0021** (0.0009)	−0.0029 (0.0032)
<i>Developing</i>	−0.2046*** (0.0327)	−0.2084*** (0.0314)	−0.1596*** (0.0267)	−0.2173*** (0.0654)
<i>Panel</i>	−0.2939** (0.1367)	−0.2916** (0.1144)	−0.3385*** (0.0937)	−0.2766 (0.1598)
<i>IV</i>	−0.0088 (0.0074)	−0.0089 (0.0073)	−0.0061 (0.0062)	−0.0113 (0.0131)
<i>GPI</i>	0.3038*** (0.0746)	0.2715*** (0.0700)	0.0861 (0.0674)	0.0381 (0.0874)
<i>Country</i>	−0.0978 (0.2002)	−0.0451 (0.1761)	0.3128** (0.1487)	0.5132 (0.4054)
<i>Industry</i>	−0.3387* (0.1767)	−0.3010** (0.1496)	−0.0994 (0.1220)	0.0113 (0.1514)
<i>Precision</i>	0.3816 (0.1931)	0.3787** (0.1610)	0.4299*** (0.1329)	0.2756 (0.2341)
<i>Constant</i>	0.0437 (1.0966)	−0.0792 (1.0648)	−1.7956* (0.9772)	−1.5283 (3.4729)
<i>N. obs. (N. Studies)</i>	256 (15)	256 (15)	246 (15)	256 (15)
<i>Log (restricted)-likelihood</i>	−585.8976	−553.2002	−516.4105	
<i>Wald test</i>	106.86***	112.23***	77.28***	
<i>R²</i>				0.2975

Notes: Standard errors for coefficient estimators are in parentheses (). Level of significance: *** for p-value<0.01; ** for p-value<0.05; * for p-value<0.1. N. Obs. is the number of observations; N. studies is the number of studies.

Table 9

Estimation of multivariate meta-regression for growth – dependent variable: t.

	Baseline Meta-regression	Robustness estimations		
		Maximum Likelihood	No outliers	OLS with clustered robust SE
	(1)	(2)	(3)	(4)
Coefficients for:				
<i>Nobs</i>	0.0081*** (0.0020)	0.0037*** (0.0009)	0.0026* (0.0014)	0.0037** (0.0012)
<i>Year</i>	0.0176* (0.0090)	0.0138*** (0.0041)	0.0130* (0.0067)	0.0138*** (0.0039)
<i>Q1Journal</i>	−0.0733 (0.0548)	−0.0719*** (0.0231)	−0.0542** (0.0261)	−0.0712* (0.0363)
<i>Ncitations</i>	0.0006** (0.0002)	0.0006*** (0.0001)	0.0005*** (0.0002)	0.0006*** (0.0002)
<i>Ncountries</i>	−0.0009 (0.0007)	−0.0009** (0.0004)	−0.0006 (0.0004)	−0.0009 (0.0006)
<i>Panel</i>	−0.1178 (0.0801)	−0.1341** (0.0520)	−0.0842 (0.0581)	−0.1341** (0.0514)
<i>GPI</i>	−0.0214 (0.0136)	−0.0237* (0.0134)	−0.0188* (0.0111)	−0.0237 (0.0171)
<i>Country</i>	0.1342 (0.0841)	0.0949** (0.0439)	0.0691 (0.0485)	0.0949 (0.0605)
<i>Industry</i>	0.1614*** (0.0366)	0.0685*** (0.0181)	0.0594** (0.0240)	0.0685** (0.0224)
<i>Precision</i>	−0.2822* (0.1695)	−0.1362* (0.0805)	−0.1752 (0.1190)	−0.1362 (0.0973)
<i>Constant</i>	0.0544 (0.8849)	−0.1070 (0.6760)	0.0964 (0.6989)	−0.1070 (0.7041)
<i>N. obs. (N. Studies)</i>	206 (12)	206 (12)	195 (12)	206 (12)
<i>Log (restricted)-likelihood</i>	−432.6628	−380.0714	−372.2307	
<i>Wald test</i>	27.16***	98.09***	25.68**	
<i>R²</i>				0.3226

Notes: See [Table 8](#).

increasing importance of IPR protection on innovation encouragement, as well as their influence on growth. Then, by means of a meta-analysis we conducted a quantitative analysis of the empirical literature that estimates the effect of the IPR on Innovation and Growth. Our approach allowed us to summarize and to systematize the empirical results of this literature and advance some explanations for the divergent results in this research field.

An important conclusion regarding the estimation of the impact of IPR on growth is that the results are influenced by publication bias, which means that editors prefer to publish studies reporting results of greater statistical significance. After correcting for this type of bias we found that the average effect of IPR on growth is positive, as is the average effect of IPR on innovation.

The meta-analysis also revealed that the effect of IPR on innovation varies with countries' development level. It is weaker in developing countries, where investment in innovative activities is low and it is preferable to imitate external innovations, than in developed countries, where firms understand more effectively the importance that IPR have on innovative activities, the capital available to invest is high, and subsidization policies have more powerful effects. We also found that the heterogeneity in the reported effects of IPR can be explained by differences in studies' methodological characteristics, such as the data structure, level of data/analysis, measurement of IPR, sample size, or publication quality.

The results of this work have relevant policy implications. For example, they imply that developing countries do not benefit significantly from policies that strengthen the degree of IPR protection. Instead, policies oriented to promote imitation of innovations from abroad could be more suitable for developing economies and contribute to rapid economic growth. On the contrary, in developed countries policies that strengthen IPR protection play a crucial role in fostering innovation and

economic growth.

The results of the meta-analysis also open some avenues for future research. In particular, they call attention to the need to explore in more detail the transmission mechanisms from IPR to innovation and from innovation to economic growth in both developed and developing countries. Moreover, given the complex nature of the R&D-innovation-IPR-growth relationships and the heterogeneity of institutional frameworks in space and time, more emphasis should be placed on estimating such relationships at a national/regional level (comparing, for example, several industries or several regions in a single country) and considering broader measures of IPR, thereby capturing dimensions other than patents.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A.1. Estimation results of the WALs routine

Table 10

Estimation of the WALs routine for innovation – dependent variable: t

	Coefficients	Standard errors	t-statistics
<i>Nobs</i>	0.0155	0.0051	3.06
<i>Year</i>	0.0037	0.0070	0.52
<i>Q1Journal</i>	−0.1000	0.0500	−2.00
<i>Ncitations</i>	−0.0002	0.0002	−0.96
<i>Ncountries</i>	−0.0023	0.0009	−2.69
<i>Developing</i>	−0.2089	0.0334	−6.25
<i>Developed</i>	0.0149	0.0354	0.42
<i>Panel</i>	−0.2328	0.0716	−3.25
<i>IV</i>	−0.0090	0.0082	−1.10
<i>Trade</i>	0.0126	0.0244	0.52
<i>GPI</i>	0.1395	0.0709	1.97
<i>Country</i>	0.1446	0.1422	1.02
<i>Industry</i>	−0.1751	0.0981	−1.78
<i>Precision</i>	0.2918	0.1292	2.26
<i>Constant</i>	0.3797	0.9347	0.41

Table 11

Estimation of the WALs routine for growth – dependent variable: t

	Coefficients	Standard errors	t-statistics
<i>Nobs</i>	0.0036	0.0014	2.60
<i>Year</i>	0.0114	0.0044	2.60
<i>Q1Journal</i>	−0.0511	0.0213	−2.40
<i>Ncitations</i>	0.0005	0.0001	3.39
<i>Ncountries</i>	−0.0007	0.0006	−1.25

(continued on next column)

Table 11 (continued)

	Coefficients	Standard errors	t-statistics
<i>Developing</i>	0.0081	0.0112	0.72
<i>Developed</i>	−0.0277	0.0366	−0.76
<i>Panel</i>	−0.1057	0.0489	−2.16
<i>IV</i>	−0.0033	0.0074	−0.45
<i>Trade</i>	−0.0074	0.0256	−0.29
<i>GPI</i>	−0.0119	0.0119	−1.00
<i>Country</i>	0.0803	0.0645	1.24
<i>Industry</i>	0.0668	0.0266	2.51
<i>Precision</i>	−0.1267	0.0870	−1.46
<i>Constant</i>	0.1309	0.7015	0.19

A.2. Estimation results of the multivariate meta-regressions with all of the covariates presented in Table 7

Table 12

Estimation of multivariate meta-regression for innovation – dependent variable: *t*

	Baseline Meta-regression	Robustness estimations		
		Maximum Likelihood	No outliers	OLS with clustered robust SE
	(1)	(2)	(3)	(4)
Coefficients for:				
<i>Nobs</i>	0.0096 (0.0056)	0.0102* (0.0058)	0.0085* (0.0049)	0.0189* (0.0101)
<i>Year</i>	0.0160 (0.0256)	0.0165 (0.0186)	0.0069 (0.1837)	0.0057 (0.0194)
<i>Q1Journal</i>	−0.1119 (0.1567)	−0.1143 (0.1143)	−0.0627 (0.1142)	−0.1168 (0.1160)
<i>Ncitations</i>	0.0002 (0.0006)	0.0002 (0.0004)	0.0002 (0.0004)	−0.0001 (0.0005)
<i>Ncountries</i>	0.0039*** (0.0012)	0.0032*** (0.0012)	0.0013 (0.0011)	−0.0029*** (0.0032)
<i>Developing</i>	−0.0951** (0.0372)	−0.1135** (0.0355)	−0.0636** (0.0304)	−0.2401*** (0.0803)
<i>Developed</i>	0.1907*** (0.0343)	0.1768*** (0.0333)	0.1687*** (0.0281)	0.0221 (0.0795)
<i>Panel</i>	−0.3722* (0.2245)	−0.3588** (0.1635)	−0.3924** (0.1625)	−0.2704 (0.1636)
<i>IV</i>	−0.0090 (0.0070)	−0.0090 (0.0068)	−0.0062 (0.0057)	−0.0140 (0.0142)
<i>Trade</i>	−0.01385 (0.0252)	−0.0170 (0.0245)	0.0109 (0.0222)	0.0032 (0.0317)
<i>GPI</i>	0.3572*** (0.0768)	0.3331*** (0.0730)	0.0920 (0.0797)	0.1543 (0.1595)
<i>Country</i>	0.6677 (0.2263)	−0.1770 (0.2041)	0.2154 (0.1941)	0.0230 (0.3506)
<i>Industry</i>	−0.2247* (0.2352)	−0.3616** (0.1807)	−0.1341 (0.1759)	−0.1331 (0.1917)
<i>Precision</i>	0.0613 (0.4367)	0.0533 (0.3207)	0.1718 (0.3134)	0.3019 (0.4825)
<i>Constant</i>	0.0460 (1.0562)	0.0841 (1.0303)	−1.7422* (0.9363)	−0.2759 (3.1968)
<i>N. obs. (N. Studies)</i>	256 (15)	256 (15)	246 (15)	256 (15)
<i>Log (restricted)-likelihood</i>	−586.6610	−539.2198	−516.7958	
<i>Wald test</i>	149.24***	156.34***	117.01***	
<i>R²</i>				0.3177

Table 13Estimation of multivariate meta-regression for growth – dependent variable: *t*.

	Baseline Meta-regression	Robustness estimations		
	(1)	Maximum Likelihood	No outliers	OLS with clustered robust SE
<i>Coefficients for:</i>				
	(1)	(2)	(3)	(4)
<i>Nobs</i>	0.0094*** (0.0021)	0.0045*** (0.00137)	0.0032 (0.0020)	0.0045 (0.0027)
<i>Year</i>	0.0176* (0.0010)	0.0136*** (0.0043)	0.0138* (0.0078)	0.0136*** (0.0032)
<i>Q1Journal</i>	−0.0753 (0.0560)	−0.0722*** (0.0232)	−0.0589 (0.0288)	−0.0722* (0.0363)
<i>Ncitations</i>	0.0006** (0.0003)	0.0006*** (0.0001)	0.0005*** (0.0002)	0.0006*** (0.0002)
<i>Ncountries</i>	−0.0009 (0.0007)	−0.0010* (0.0005)	−0.0006 (0.0006)	−0.0009 (0.0008)
<i>Developing</i>	0.0036 (0.0136)	0.0074 (0.0133)	0.0021 (0.0011)	0.0075 (0.0201)
<i>Developed</i>	−0.0536 (0.0465)	−0.0454 (0.0461)	−0.0471 (0.0381)	−0.0454 (0.0771)
<i>Panel</i>	−0.1001 (0.0891)	−0.1251** (0.0529)	−0.0773 (0.0648)	−0.1251** (0.0536)
<i>IV</i>	−0.0034 (0.0091)	−0.0032 (0.0089)	−0.0048 (0.0076)	−0.0033 (0.0056)
<i>Trade</i>	−0.01165 (0.0471)	−0.0079 (0.0251)	0.0003 (0.0308)	−0.0079 (0.00315)
<i>GPI</i>	−0.0201 (0.0140)	−0.0219 (0.0136)	−0.0168 (0.0114)	−0.02191 (0.01482)
<i>Country</i>	0.1279 (0.0899)	0.1032 (0.0.064)	0.0613 (0.0670)	0.1032 (0.0976)
<i>Industry</i>	0.1708*** (0.0380)	0.0818*** (0.0259)	0.0671*** (0.0033)	0.0818 (0.0512)
<i>Precision</i>	−0.3060* (0.1861)	−0.1552* (0.0894)	−0.2072* (0.1361)	−0.1552 (0.0872)
<i>Constant</i>	−0.3316 (0.9662)	0.0720 (0.7763)	0.3356 (0.8091)	0.0720 (0.7978)
<i>N. obs. (N. Studies)</i>	206 (12)	206 (12)	195 (12)	206 (12)
<i>Log (restricted)-likelihood</i>	−443.3193	−379.2002	−383.7503	
<i>Wald test</i>	28.38**	100.67***	24.54**	
<i>R²</i>				0.3283

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